Completing the dihedral attachment branch

Human Compression project: bounded L044 proof component

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le A_n$, $n \ge 12$, be a transitive string C-group whose maximal omissions are intransitive. Suppose a unique eligible 2-edge joins the G_2-orbits $A = \{1, \dots, 5\}$ and X, with

$$\rho_0 = l = (1\ 2)(3\ 4), \qquad \rho_1 = s = (2\ 3)(4\ 5).$$

Assume $K = G_{>2}$ is supported on and transitive on X; write $q = |X|$ and $d = r-3$. The crossing endpoint in A is necessarily 5, the sole fixed point of l. We assume the alternating rank bound at every strictly smaller degree at least 12. The accepted proper-primitive, anchored-imprimitive and small anchored-interval results are paid supporting inputs, not a blanket unanchored rank bound.

Proposition 1. *Either $r \le \lfloor (n-1)/2 \rfloor$, or the actual prefix $\langle \rho_0, \rho_1, \rho_2 \rangle$, with its ordered generators, is exactly source Table 2 row (2) or row (3) on six points and fixes the remaining points. The dual statement holds.*

Proof. The tail K has a genuine even string C-marking. Every maximal omission in K is intransitive on X: otherwise adjoining the head, transitive on A, and the crossing involution would make the corresponding global omission transitive. All its generators except the first fix the crossing endpoint in X, by far commutation. Its diagram is connected, since a commuting marked component away from that endpoint would fix it and hence, by transitivity and commutation, fix all of X.

If $q < 12$ and $d \ge (q-1)/2$, the accepted small primitive and anchored-imprimitive interval results leave only source Table 2 rows (1)–(3). Their support degrees are five or six, which would give $n \le 11$. Thus $d < (q-1)/2$ for $q < 12$. For $q \ge 12$, the accepted proper-primitive and anchored- imprimitive bounds give $d \le \lfloor (q-2)/2 \rfloor$ unless $K = \mathrm{Alt}(X)$. In all these cases

$$r = d + 3 \le \left\lfloor \frac{q+4}{2} \right\rfloor = \left\lfloor \frac{n-1}{2} \right\rfloor.$$

In the remaining case the explicit smaller-degree induction hypothesis gives $d \le \lfloor (q-1)/2 \rfloor$. A failure of the asserted bound therefore requires

$$q \ge 13 \text{ odd}, \qquad K = \mathrm{Alt}(X), \qquad d = (q-1)/2.$$

Name the crossing endpoint in X as 6. The unique cross transposition and $[l, \rho_2] = 1$ give

$$\rho_2 = (5\ 6)uv,$$

where u is an involution in $C_{S_{\{1,2,3,4\}}}(l)$, and v is an involution on $X \setminus \{6\}$. Their signs are opposite. The centralizer preserves the two pairs of l and has order eight; its involutions, including the identity, are exactly

$$1,\ (1\ 2),\ (3\ 4),\ (1\ 2)(3\ 4),\ (1\ 3)(2\ 4),\ (1\ 4)(2\ 3).$$

We dispatch all six literal choices, without replacing a marked tuple by an abstract group of the same order.

If $u = 1$, then $R = \langle \rho_2, K \rangle$ is an actual even subgroup supported on $X \cup \{5\}$. It is the full $\mathrm{Alt}(X \cup \{5\})$, by overlapping natural alternating supports. Its full IP is inherited from G, since no projection was taken. Its rank is $d + 1 = (q + 1)/2$ and its degree $q + 1$ is strictly smaller than $n = q + 5$ and at least 14. This contradicts the stated induction hypothesis.

If $u = (1\ 2)(3\ 4)$ or $u = (1\ 3)(2\ 4)$, the edges of s and u connect all five points of A. Together with K on X and the crossing edge, the actual subgroup $G_0 = \langle s, \rho_2, K \rangle$ is transitive on Ω. This is forbidden. For the first choice the connecting path is $1, 2, 3, 4, 5$; for the second it is $1, 3, 2, 4, 5$.

If $u = (1\ 4)(2\ 3)$, conjugate by the permutation $(1\ 4\ 2\ 3)$, fixing 5 and X. The first three marks become

$$(1\ 2)(3\ 4), \quad (2\ 5)(1\ 3), \quad (1\ 3)(2\ 4)(5\ 6)v.$$

The complete dihedral–alternating obstruction applies and excludes this case. In particular the decorated six-point S_5 possibility is not silently identified with either of the source's A_5 rows.

It remains to take $u = (1\ 2)$ or $u = (3\ 4)$. Projection of $P = \langle l, s, \rho_2 \rangle$ to $\{1, \ldots, 6\}$ has the literal generators of accepted source Table 2 row (2) or row (3), respectively. Both are the perfect group A_5 in that precise degree-six action. The exterior projection of P lies in $\langle v \rangle$. If $v \neq 1$ and the six-point projection were injective, the exterior action would define a nontrivial homomorphism $A_5 \to C_2$. Consequently $v \in P$. But v is even, so belongs to $K = \mathrm{Alt}(X)$, while full ambient IP gives $P \cap K = 1$. Thus $v = 1$. The two remaining actual ordered prefixes are exactly the two stated source rows. Reversing the full string gives the dual conclusion. □

Source, dependencies and correction scope. This is the complete dihedral attachment argument for Cameron–Fernandes–Leemans–Mixer, *Highest rank of a polytope for A_n*, arXiv:1605.09173v1, Proposition 5.14, physical pages 28–29, in the supported interval setting of its Section 5.2. It reuses the source's rank-deficit and sesqui-extension strategy. The six-choice normalization is stated explicitly and the previously uncovered S_5 decoration is excluded by the new full-IP obstruction. The original conclusion is retained; no counterexample to that conclusion is asserted.

Both source rows and their exact A_5 identifications are already covered by the accepted Table 2 packet; its complete required census and source transcription remain paid inputs. No census is repeated here. The alternating rank step is an explicit strong-induction hypothesis, not a claim that all remaining L044 branches have already been proved. The exceptional prefixes, their later source cases, the connected-square branch and the full source assembly remain separate. No new formal, original or question credit follows from this component.